IMPERIAL

From Deterministic Forecasts to Probabilistic Inference With Approximate Bayesian Computation

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1. Predictive Distributions From Point Forecasts

Let $\mathcal{X} \subseteq \mathbb{R}^d$ denote the state space. A point forecast $\tilde{x}_{t+1}$ given by a deterministic forecasting function $f : \mathcal{X} \to \mathcal{X}$ $\tilde{x}_{t+1} = f(x_t)$, does not capture uncertainty. In applications, predictive distributions expressing uncertainty are typically required. This means that we are concerned with the conditional distribution

$$\tilde{\pi}(x_{t+1} \mid x_t),$$

which can be obtained by introducing a stochastic perturbation parametrised by $\theta \in \Theta \subseteq \mathbb{R}^p$:

$$\tilde{\pi}_\theta(x_{t+1} \mid x_t).$$

Example: AR(1) with noise

$$x_{t+1} = \phi x_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(\mu, \sigma^2)$$
$$\Rightarrow \tilde{\pi}_{\mu,\sigma^2}(x_{t+1} \mid x_t) = \mathcal{N}(\phi x_t + \mu, \sigma^2)$$

In order to conduct posterior inference, we need a method to infer sensible stochastic schemes to apply to a given deterministic model.

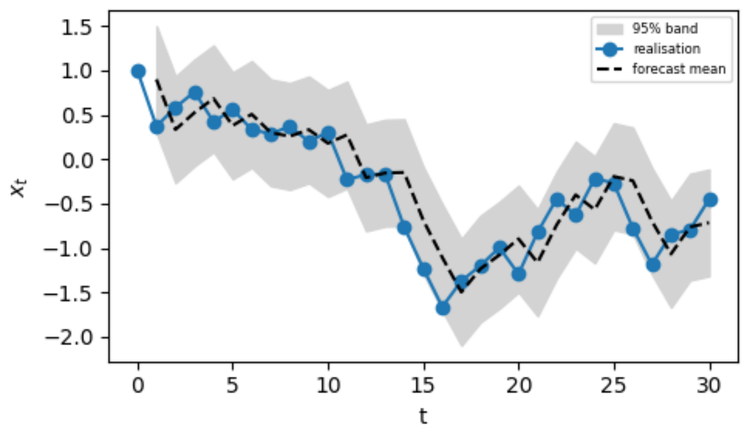


Figure 1: AR(1) forecasts and realisations with Gaussian noise

In this case, a conditional predictive distribution is available in closed form. However in many practical scenarios, closed-form expressions are rarely available...

2. Intractable Likelihoods

In weather-forecasting systems, as well as numerous other settings, the predictive distribution $\tilde{\pi}_\theta(x_{t+1} \mid x_t)$ is only implicitly defined. The likelihood

$$L(\theta) = \prod_{t=0}^{T-1} \tilde{\pi}_\theta(x_{t+1} \mid x_t)$$

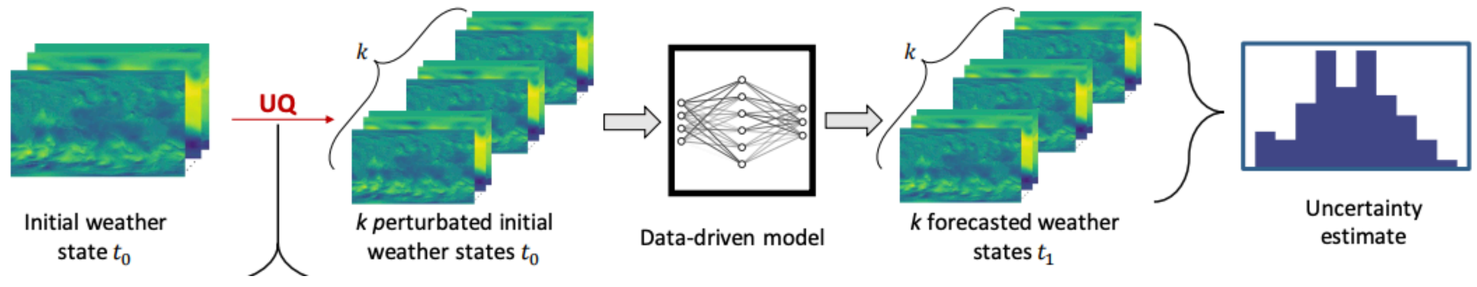


Figure 2: Bülte et al. 2025: Posterior approximations arising from initial condition perturbations applied to machine-learning weather prediction

cannot be directly examined, rendering standard Bayesian methods such as MCMC and variational methods inapplicable.

3. Forecast Properties as Constraints

Injecting unmotivated noise into a deterministic forecast may produce baseless predictive distributions that are inadequate for applied forecasting situations. Forecasts should instead satisfy statistically and practically meaningful properties. Examples of such properties include:

- **Reliability**: forecast probabilities align with observed event frequencies;
- **Resolution**: forecasts are meaningfully sensitive to the input state x_t;
- **Uncertainty**: the marginal forecast distribution expresses genuine probabilistic entropy.

The chosen criteria define an admissible parameter set

$$\mathcal{C} = \{\theta \in \Theta : \tilde{\pi}_\theta \text{ satisfies forecast criteria}\}$$

resulting in a constrained posterior

$$P(\theta \mid \mathcal{C}, \mathcal{D}), \quad \mathcal{D} = \{(x_t, x_{t+1})\}_{t=0}^{T-1}.$$

Score functions

Posterior compliance can be practically assessed via forecast–outcome pairs $(\tilde{\pi}_\theta(\cdot \mid x_t), x_{t+1})$. A scoring function

$$S : \mathcal{P}(\mathcal{X}) \times \mathcal{X} \to \mathbb{R}$$

assigns a real-valued penalty to the forecast $\tilde{\pi}_\theta(\cdot \mid x_t)$ evaluated against the realised outcome x_{t+1}.

Define the aggregated forecast score under parameter θ as the empirical average

$$\hat{S}(\tilde{\pi}_\theta, \mathcal{D}) := \frac{1}{T} \sum_{t=0}^{T-1} S\big(\tilde{\pi}_\theta(\cdot \mid x_t), x_{t+1}\big).$$

Given a tolerance threshold $\epsilon \in \mathbb{R}$, the admissible parameter set is thus

$$\mathcal{C} = \{\theta \in \Theta : S(\tilde{\pi}_\theta, \mathcal{D}) < \epsilon\}$$

Example: Probabilistic Binary Variable Forecasting

For Binary variables, the Brier score $\frac{1}{T}\sum_t^T (\tilde{\pi}(\tilde{x}_{t+1} = 1 | x_t) - x_t)^2$ is a natural way to assess the reliability, resolution and uncertainty of probabilistic predictions.

4. Approximate Bayesian Computation

ABC produces a likelihood-free approximation to $P(\theta \mid \mathcal{C}, \mathcal{D})$ via simulation and discrepancy rejection.

1. **Propose** Sample $\theta \sim q(\theta)$, a proposal distribution.
2. **Simulate** Generate a synthetic sample from the proposed forecasting distributions. $\mathcal{D}^{(\theta)}$ from $\tilde{\pi}_\theta$.
3. **Discrepancy** Compute the proposal's forecast score: $\Delta(\theta) = \hat{S}(\tilde{\pi}_\theta, \mathcal{D})$
4. **Accept** Keep θ if $\Delta(\theta) \leq \varepsilon$, forming $\hat{\mathcal{C}}(\mathcal{D})_\varepsilon = \{\theta : \Delta(\theta) \leq \varepsilon\}$.
5. **Weight** Importance–correct for a generic proposal: $P_\varepsilon(\theta \mid \mathcal{D}) \propto \frac{\tilde{\pi}(\theta)}{q(\theta)} \mathbf{1}_{\hat{\mathcal{C}}(\mathcal{D})_\varepsilon}$.

Which approximates $P(\theta \mid \hat{\mathcal{C}}(\mathcal{D}), \mathcal{D})$ as $\varepsilon \to 0$ and $q \equiv \pi$.

5. Research Objectives

- **Algorithmic development**: establish ABC schemes for high-dimensional θ under forecast property constraints.
- **Application**: assess framework viability and usefulness in ML-based weather prediction.

References

(Bülte et al. 2025; Siegert 2017; Clarté et al. 2021; Beaumont 2019)

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